Euler's Method and Logistic Growth

Section 1: Euler's Method Iterations

1 Given $\frac{dy}{dx} = x + y$ and initial condition $y(0) = 1$, use Euler's method with step size $\Delta x = 0.5$ to approximate $y(1)$.

a. 2.25

b. 2.00

c. 2.50

d. 2.75

2 If $y'' > 0$ (concave up) on the interval, does Euler's method yield an overestimate or underestimate of the true solution?

a. An underestimate because tangent segments lie below the curve

b. An overestimate because tangent segments lie above the curve

c. Exact value

d. Depends on the step size

3 Given $\frac{dy}{dx} = 2x$ with $y(1) = 3$, what is the error in approximating $y(2)$ using Euler's method with $\Delta x = 0.5$?

a. 0.5

b. 1.0

c. 0

d. 0.25

Section 2: The Logistic Differential Equation

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4 A population grows according to $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$. What is the carrying capacity of this population?

a. 500

b. 0.04

c. 250

d. $12,500$

5 For the logistic model $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$, at what population size is the growth rate $\frac{dP}{dt}$ maximized?

a. 250

b. 500

c. 125

d. 375

6 What is $\lim_{t \to \infty} P(t)$ for a logistic population with carrying capacity 800 if $P(0) = 50$?

a. 800

b. ∞

c. 400

d. 50

Section 3: Analytical Solutions to Logistic Models

7 What is the general form of the solution to the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{M})$?

a. $P(t) = \frac{M}{1 + Ae^{-kt}}$

b. $P(t) = Me^{kt}$

c. $P(t) = M - Ae^{-kt}$

d. $P(t) = \frac{M}{1 - Ae^{kt}}$

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8 If $P(t) = \frac{1000}{1 + 9e^{-0.2t}}$, what is the initial population $P(0)$?

a. 100

b. 1000

c. 90

d. 10

9 What is the concavity of the logistic curve when $P(t) > \frac{M}{2}$?

a. Concave down ($P''(t) < 0$)

b. Concave up ($P''(t) > 0$)

c. Zero concavity

d. Undefined

Section 4: Advanced Slope Fields with Autonomous Equations

10 In the autonomous differential equation $\frac{dy}{dx} = y(y - 3)(y + 2)$, what are all equilibrium solutions?

a. $y = 0, y = 3, y = -2$

b. $y = 3, y = -2$ only

c. $y = 0$ only

d. $x = 0, x = 3, x = -2$

11 For $\frac{dy}{dx} = 2y(1 - \frac{y}{4})$, which equilibrium solution is stable (an attractor)?

a. $y = 4$

b. $y = 0$

c. Both are stable

d. Neither is stable

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12 If $y(0) = 6$ in $\frac{dy}{dx} = 2y(1 - \frac{y}{4})$, what is $\lim_{x \to \infty} y(x)$?

a. 4

b. 6

c. ∞

d. 0

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